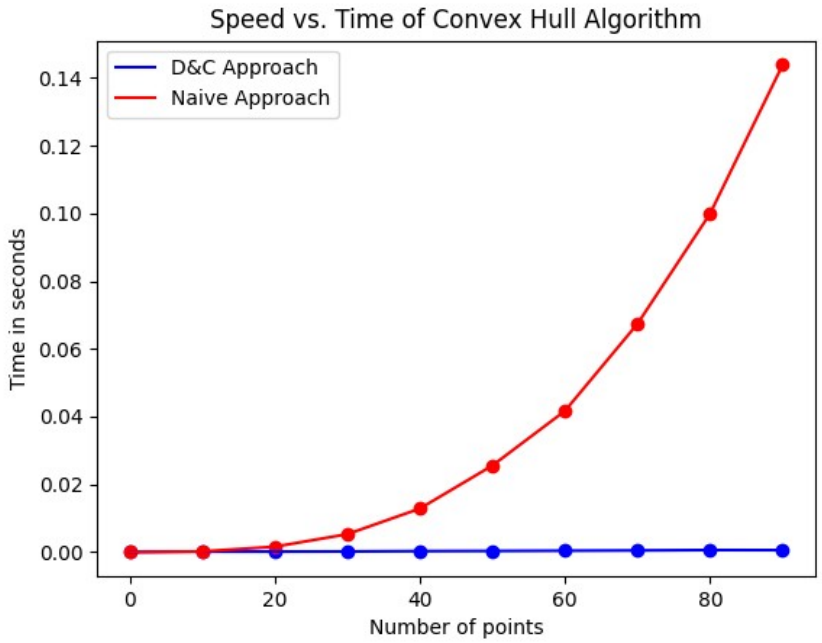
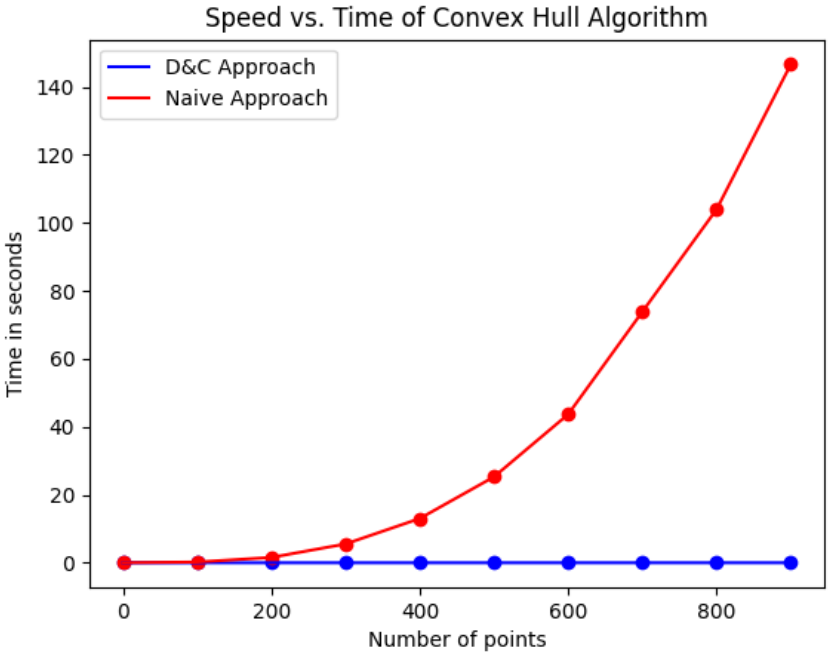
Convex Hull Benchmarking

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The ‘naive algorithm’, implemented in base_case_hull, performed basically as one would expect. Since the worst-case scenario is that each point must be compared to each other point, the algorithm has a theoretical runtime of $O(n^2)$. The behavior found during testing matches this expectation, as you can see along the red curve of both graphs.

The ‘divide-and-conquer’ algorithm also performs as expected, and you can see that the difference in performance between the two algorithms is drastic for larger inputs. The theoretical runtime of this approach is O(n), and this linear runtime is also clearly visible in the blue ‘curve’ of the both graphs. The blue data is linear, indicating that the actual runtime matches our theoretical expectations.





You can also see that the two graphs appear nearly identical in shape, despite the data used to create each being different by an order of magnitude. The same pattern is also observable when looking at the raw data, where divide-and-conquer time goes up linearly, about ½ thousandth of a second per 100 points, whereas the naive time shoots up at a roughly quadratic rate.

Number of points	D&C time taken (seconds)	Naive time taken (seconds)
0	0.000	0.000
10	0.000	0.000
20	0.000	0.002
30	0.000	0.005
40	0.000	0.013
50	0.000	0.026
60	0.000	0.042
70	0.000	0.067
80	0.001	0.100
90	0.001	0.144

Number of points	D&C time taken (seconds)	Naive time taken (seconds)
0	0.000	0.000
100	0.001	0.210
200	0.001	1.567
300	0.002	5.526
400	0.002	13.018
500	0.003	25.279
600	0.004	43.679
700	0.005	73.873
800	0.005	104.000
900	0.006	146.663